

Dual Adversarial Adaptation With Dynamic Labeling Mechanism for Semisupervised Cross-Well Lithology Identification

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Abstract—Well log instruments continuously measure multiple geophysical parameters of wellbore rocks, providing high-resolution observation data for identifying subsurface lithology. Recently, lithology identification methods based on semisupervised learning and transfer learning have been investigated to address the widespread issue of label scarcity. However, these methods only account for the distribution discrepancy between wells (domains), while ignoring the data divergence within a well. This can result in inaccurate pseudolabels within a well and erroneous label matching across wells. To cope with these challenges, we propose a dual adversarial adaptation method with dynamic labeling mechanism for semisupervised cross-well lithology identification. Our method leverages intradomain adversarial learning to promote fine-grained category alignment and interdomain adversarial learning to generate domain-invariant feature representations, thereby reducing the data differences within and between wells. Additionally, our method introduces a dynamic label adjustment mechanism to optimize the selection of target pseudolabels and source samples, thereby increasing reliable supervision information and improving the feature alignment process. Experimental results on actual well log datasets demonstrate that our method can achieve effective cross-well lithology identification when target labels are scarce.

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Impact Statement—Lithology identification from well log curves is crucial for constructing fine-grained geological models. However, existing studies face challenges in predicting lithology across wells due to the combined effects of limited labels and distribution differences. We propose to perform adversarial learning both across and within wells, combined with dynamic labeling assignment, to overcome these challenges. Experiments on real-world well log datasets demonstrate that the proposed method improves the macroaverage recall of cross-well lithology identification by 12.6% on average compared to existing methods, when only 10% of the labels are available for the target well. This advance provides a feasible solution for geological modeling under scarce labels and cross-scenario conditions.

Index Terms—Adversarial learning, domain adaptation, lithology identification, semisupervised learning, well logs.

I. INTRODUCTION

WELL log instruments record the geophysical properties (such as acoustics, electricity, and nuclear magnetism) of rocks surrounding the wellbore, enabling continuous characterization and high-resolution modeling of the borehole geology [1], [2], [3]. As a fundamental task in geological modeling, lithology identification aims to interpret and infer the depth-wise lithology category using multiple well log curves measured by these instruments [4]. Early studies on well log-based lithology identification employed empirical formulas to infer lithology by developing shale content equations [5] or cross-plots [6]. These methods are adept at handling linear relationships between a small number of strongly correlated well logs and lithology. However, they encounter challenges in modeling the dependencies among multidimensional well logs, which limits their accuracy in interpreting complex reservoirs.

Since the well log curves are well structured and have strong regularity, machine learning methods have recently shown great potential in lithology identification [7]. Numerous supervised learning-based lithology identification methods have been proposed, evolving from classical machine learning to powerful ensemble learning and deep learning, and from simple application to insightful feature representation [8], [9], [10], [11]. For example, Zhou et al. [12] employ SMOTE to address the imbalance among lithology classes and develop a lithology identification model based on the gradient boosting decision trees (GBDT). Considering the variation shape of well logs, Li et al. [13] apply

the local mean decomposition to extract multidepth shape features and perform multigrained scanning on the optimized well log features to enhance the feature representation. Zhang et al. [14] proposed a sequence correlation filtering method based on one-dimensional convolution, which leverages category-wise filtering to improve the discriminability of different lithology types within logging features. Other approaches combine well logs and seismic data for lithofacies prediction [15], [16].

Despite the progress of supervised learning methods, they rely heavily on sufficient lithology labels, which are difficult to obtain in practice due to the high cost of coring operations. Researchers have shifted their focus to lithology identification under the more practical issue of label scarcity. Some semisupervised lithology identification methods have been proposed to assign pseudolabels to unlabeled samples. For example, Dunham et al. [17] apply label propagation and self-training to enrich the number of lithology labels and improve supervised learning. Zhao et al. [18] use a predefined threshold outside the training loop to generate pseudolabels. Nevertheless, existing methods are not able to dynamically adjust the threshold of pseudolabels with the learning progress. This static label assignment can lead to inaccurate pseudolabels when there are distribution differences between labeled and unlabeled samples, posing a great challenge for semisupervised lithology identification.

In addition to semisupervised learning, some studies have proposed leveraging auxiliary domains for transfer learning to tackle the issue of insufficient labels on the target well. In this paradigm, wells with sufficient labeled samples (interpreted wells) are regarded as the source domain, while wells with only a few or no labeled samples (uninterpreted wells) are regarded as the target domain [19]. Due to the differences in well log instruments and measurement technologies, data from different wells (domains) have varying distributions [20], [21]. To this end, domain adaptation methods have been proposed to align feature distributions and reduce distribution discrepancies. For example, Ren et al. [22] use the joint maximum mean discrepancy (MMD) to learn transferable and discriminative features of well logs. Xu et al. [23] combine contrastive learning and adversarial learning to improve the generalization of the lithology identification model. Considering the physical meaning of well log variables, Wu et al. [24] employ the projected MMD to reduce the distribution discrepancy of well logs in a unilateral alignment way.

Although existing methods have considered the cross-well data distribution discrepancy, they neglect the divergence of data within the same well. In fact, due to differences in sedimentary environments and fluid characteristics of different strata, the well log data on the same well may be divergent. As shown in Fig. 1(a), we can observe significant distribution discrepancy between interpreted wells and uninterpreted wells. Beyond the interwell discrepancy, in Fig. 1(c), there are also distribution differences between labeled and unlabeled samples on a single well, i.e., intrawell divergence. However, existing methods did not take into account the impact of intrawell divergence on domain alignment, which can lead to erroneous category

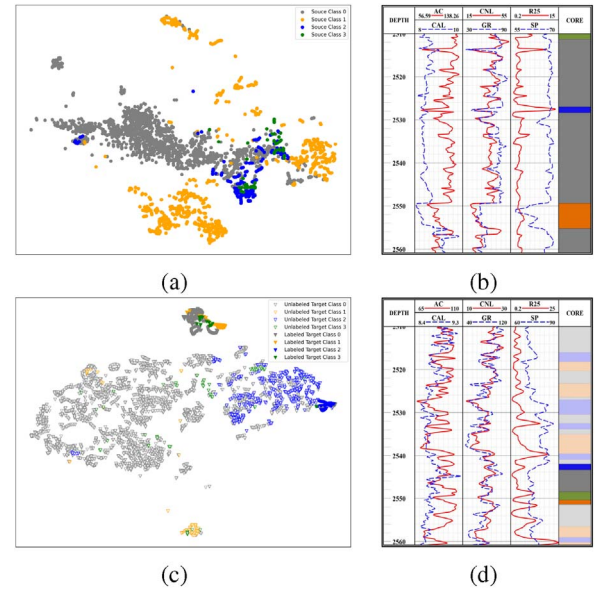


Fig. 1. Interwell discrepancy and intrawell divergence of well log data distributions. (a) and (c) t-SNE visualization of samples from wells A and B. (b) Data from well A are fully labeled, where CORE denotes the ground-truth lithology from core sampling. (d) Data from well B have only a few labels.

matching and unstable training process, posing another challenge for cross-well lithology identification.

To address the above challenges, this study proposes a dual adversarial adaptation method with dynamic labeling mechanism for semisupervised cross-well lithology identification. Specifically, we establish a dual adversarial architecture to reduce the intrawell divergence and interwell discrepancy. In the target intradomain adversarial adaptation, we use multiple class-specific discriminators to promote class-wise alignment within the target domain, making the domain structure more compact. In interdomain adversarial adaptation, we use a domain discriminator to enforce the learning of domain-invariant features, thereby aligning the feature distribution between the source and target domains. Furthermore, to fully leverage the target label information and improve the feature alignment process, we propose a dynamic label adjustment mechanism that adaptively optimize the selection of target pseudolabels and relevant source samples. Through the above dual adaptation and label adjustment, our method can achieve accurate cross-well lithology identification. The proposed method is evaluated on actual well log data from an oilfield in eastern China. Comparison results and feature visualization demonstrate the effectiveness of our method in cross-well lithology identification. The contributions of this study are summarized as follows.

- 1) We propose a dual adversarial adaptation method to reduce the intrawell divergence and interwell discrepancy of well log data.
- 2) We propose a dynamic label adjustment mechanism to increase reliable supervision information and improve the training process.

- 3) Extensive experiments on actual well log datasets demonstrate that our method can achieve effective cross-well lithology identification when target labels are scarce.

The rest of this article is organized as follows. First, we introduce the related work. Second, we present a detailed exposition of the proposed semisupervised cross-well lithology identification method. Then, we describe the experimental settings and analyze the comparative results. Finally, we present the conclusion of our work.

II. RELATED WORK

A. Semisupervised Lithology Identification

Label scarcity is a common problem in well log interpretation, leading to reduced generalization ability of supervised machine learning methods. Researchers have recently attempted to introduce semisupervised learning approaches to address this problem. Lan et al. [25] introduced positive and unlabeled learning, gradually restoring the pseudolabels of the unlabeled portion with a fixed threshold to identify different logfacies with limited positive labels. Li et al. [26] applied label propagation algorithms to assign pseudolabels and further used decision tree to generate ensemble pseudolabels of lithology categories. Zhu et al. [27] separated the well-logging curves as different feature-vision, and then applied trilearning to produce ensemble pseudolabels. Hong et al. [28] combined active learning with semisupervised learning, using informative queries to promote a self-train pseudolabel framework. Ren et al. [29] introduced pseudolabel detection based on entropy and probability criteria to provide a global threshold for pseudolabels to utilize unlabeled well-logging samples. Dong et al. [30] introduced a semisupervised ladder network, which combines unsupervised denoising autoencoder for unlabeled data, and then selects the top several pseudolabel samples to adopt by confidence level. However, existing semisupervised methods usually use a fixed threshold to generate pseudolabels, which could lead to inaccurate pseudolabels and thus limit the performance of the model, especially when there are distribution differences between labeled and unlabeled data [31]. In contrast, our method enables the model to dynamically adjust the confidence thresholds according to its learning progress, thereby improving the accuracy of the model.

B. Cross-Domain Lithology Identification

Domain adaptation methods have recently been applied in lithology identification to address the problem of distribution discrepancy of well log data caused by differences in logging instruments and measurement technologies [20]. For instance, Zhou et al. [32] proposed a streamlined pretraining and self-training framework and selected high-confidence lithology hard labels for each class in the target domain based on the proportion of classes from the source domain. Wu et al. [24] used class-aware maximum mean discrepancy (MMD) as an explicit measure of domain differences and introduced a one-side alignment module to retain source domain information while combating data drift. Ren et al. [22] addressed the domain

differences by using a denoising autoencoder to reconstruct differences and align the feature spaces. The class-aware MMD and maximized MMD are employed to preserve separability and enhance model discriminability. Although existing studies have considered the problem of cross-well distribution discrepancy, most of them ignore the data divergence within the well, which can affect the matching of categories and the stability of cross-domain training. In this article, we propose a dual adversarial adaptive method to simultaneously address the interwell discrepancy and intrawell divergence.

C. Semisupervised Domain Adaptation

Domain adaptation aims to narrow the discrepancy between different domains. Semisupervised domain adaptation (SSDA) has made recent progress, with research related to this article primarily focusing on addressing intradomain gaps and improving pseudolabel accuracy. Li et al. [33] propose an adversarial adaptive clustering method that groups samples with similar features into clusters to enhance intraclass compactness. Pan et al. [34] devise an entropy-based ranking strategy to partition target samples into easy and hard subdomains, and employ an intradomain discriminator to reduce the gap between them. Different from these methods, our study considers the distribution differences between labeled and unlabeled data in the target domain, and designs class-specific discriminators to mitigate the intradomain differences. Li et al. [35] construct a heterogeneous graph to capture cross-domain sample associations and propose a pruning strategy based on confidence uncertainty and prediction dissimilarity to reduce erroneous pseudolabels. He et al. [36] leverage label consistency across different clustering algorithms to improve pseudolabel accuracy. Huang et al. [37] enhance pseudolabel reliability through probability-level consistency and historical prediction regularization. Zhang et al. [38] investigate a more general SSDA setting, refining pseudolabels by balancing their distribution. In contrast to these methods, which generally adopt static or global expectation-based thresholds to determine high-confidence pseudolabels, our study incorporates both global and class-specific prediction status to dynamically optimize the confidence thresholds.

III. METHODOLOGY

A. Overview

First, we present the problem formulation of semisupervised cross-well lithology identification. Suppose there are ν types of logging curves. A training sample is a ν -channel sequence of length H , denoted as $\mathbf{x} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(\nu)}] \in \mathbb{R}^{H \times \nu}$, and its label y is the lithology at the middle depth of $\mathbf{x}$. Denote the data from the interpreted well as the source domain $\mathcal{D}_s = (X_s, Y_s) = \{(\mathbf{x}_i^s, y_i^s)\}_{i=1}^{n_s}$, where n_s is the number of source samples, $y_i^s \in \Omega = \{1, 2, \dots, M\}$, Ω is the label space, and M is the total number of lithology categories. Denote the data from the uninterpreted well as the target domain $\mathcal{D}_t = \mathcal{D}_{tl} \cup \mathcal{D}_{tul}$ which contains a small number of labeled samples $\mathcal{D}_{tl} = (X_{tl}, Y_{tl}) = \{(\mathbf{x}_i^{tl}, y_i^{tl})\}_{i=1}^{n_{tl}}$ and a large number of unlabeled samples $\mathcal{D}_{tul} = X_{tul} = \{\mathbf{x}_i^{tul}\}_{i=1}^{n_{tul}}$, where $n_{tl} \ll n_s$ and

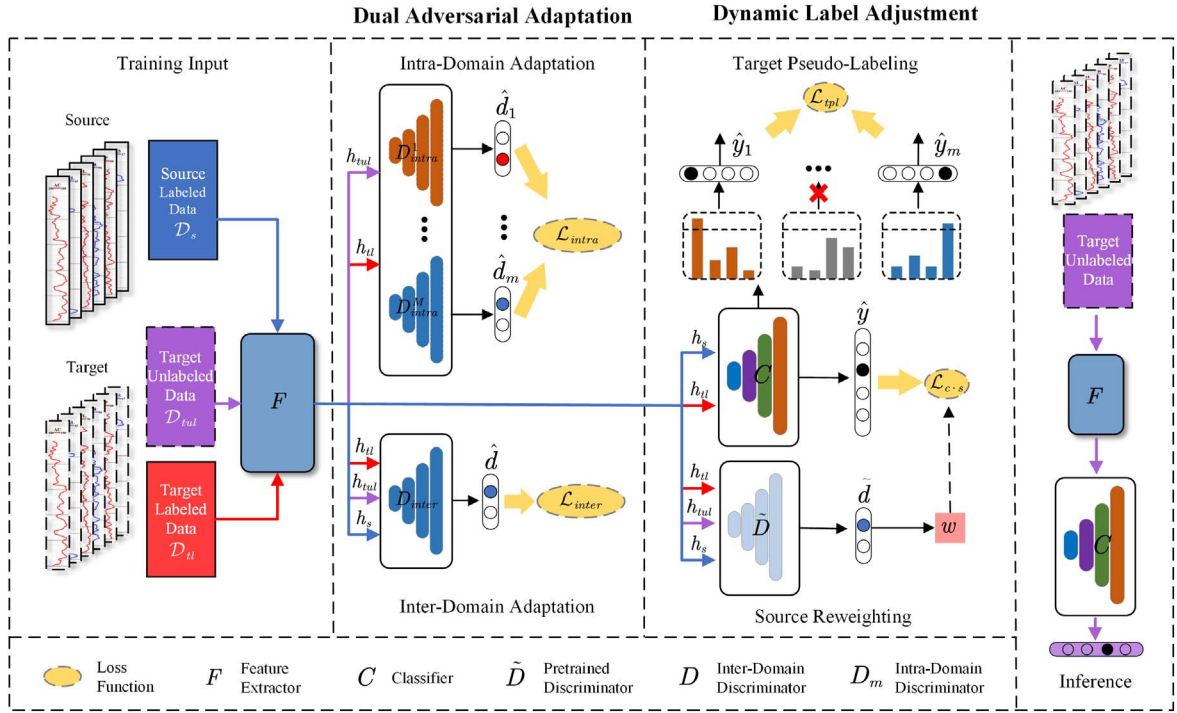


Fig. 2. Framework of dual adversarial adaptation method with dynamic labeling mechanism for semisupervised cross-well lithology identification.

$n_{tl} \ll n_{tul}$. Typically, $\mathcal{D}_s$ and $\mathcal{D}_t$, as well as $\mathcal{D}_{tl}$ and $\mathcal{D}_{tul}$ obey different marginal probability distributions. Our goal is to predict the lithology label of $\mathcal{D}_{tul}$ using $\mathcal{D}_s$ and $\mathcal{D}_t$.

To achieve this goal, we propose a dual adversarial adaptation method with dynamic labeling mechanism, as shown in Fig. 2. The proposed method consists of two major modules: a dual adversarial adaptation module and a dynamic label adjustment module. The former includes intradomain adversarial learning to facilitate fine-grained category alignment and interdomain adversarial learning to generate domain-invariant feature representations. The latter includes target pseudolabeling to optimize the selection of reliable pseudolabels and source reweighting to emphasize source samples similar to the target domain. Through these modules, our method reduces data differences and enhance feature alignment, enabling accurate cross-well lithology identification.

The specific network structure is as follows. First, the labeled source data $\mathcal{D}_s$, the unlabeled target data $\mathcal{D}_{tul}$, and the labeled target data $\mathcal{D}_{tl}$ are fed into a shared feature extractor F , producing intermediate features h_s , h_{tul} , and h_{tl} , respectively (as shown in Fig. 2). F is composed of one-dimensional convolutional layers and channel attention modules to capture both the shape and fine-grained features of well log curves [39]. Subsequently, the intermediate features are processed by four branches.

- 1) h_{tul} and h_{tl} are input into class-specific discriminators D_{intra}^m to reduce intrawell divergence.
- 2) h_s , h_{tul} , and h_{tl} are fed into a domain discriminator D_{inter} to reduce interwell discrepancy.
- 3) h_s , h_{tul} , and h_{tl} are further passed to another domain discriminator $\tilde{D}$ to evaluate the similarity between source and target samples.

- 4) h_s and h_{tl} are input into a classifier C , which incorporates a target pseudolabeling strategy to facilitate lithology classification.

Following the configurations of existing domain adaptation studies [40], all discriminators D and the classifier C are implemented as fully connected layers with different initialization parameters. The following sections will provide a detailed description of each module.

B. Dual Adversarial Adaptation

To eliminate the intrawell divergence and interwell discrepancy, we expect to compact the class-specific information within the domain and extract transferable knowledge between domains. Following the recent popular adversarial domain adaptation [40], [41], [42], [43], [44], we employ the adversarial learning paradigm to optimize the domain discriminators and feature extractors. A dual adversarial adaptation architecture is proposed to promote the feature compaction and transfer. We present a detailed exposition of the two adversarial approaches below.

1) *Intradomain Adversarial Adaptation:* In practical production, a single wellbore drilling operation spans various strata and different sedimentary environments, leading to feature divergence among samples collected from spatially distant locations. Due to the extremely limited number of labeled samples available in the target well, there are inevitably distribution differences between labeled and unlabeled data. Such intrawell divergence can lead to category mismatching during the feature alignment.

To address this problem, an intradomain adversarial adaptation module is designed to achieve class-wise alignment, as

shown in Fig. 2. Specifically, the class-specific discriminators, denoted as D_{intra}^m , $m = 1, 2 \dots M$, are designed for each lithology category to match the labeled and unlabeled data associated with category m in the target domain. Since the class of unlabeled data is unknown, it is necessary to determine which class discriminator should be assigned to each unlabeled sample. Considering that the output of the classifier $p_i = C(\mathbf{x}_i)$ can indicate the probability of the sample $\mathbf{x}_i$ belonging to each class, we use p_i as the attention coefficient to weight the feature of $\mathbf{x}_i$. Then, the process of intradomain adversarial learning can be formulated as follows:

$$\begin{aligned} \mathcal{L}_{d\text{-intra}}(\vartheta_f, \vartheta_{d\text{-intra}}) = \\ - \frac{1}{n_{tl} + n_{tul}} \sum_{m=1}^M \sum_{\mathbf{x}_i \in \mathcal{D}_t} \mathcal{L}_e(D_{\text{intra}}^m(p_i^m F(\mathbf{x}_i; \vartheta_f); \vartheta_{d\text{-intra}}), e_i) \end{aligned} \quad (1)$$

where $\mathcal{L}_e$ is the cross-entropy loss for class-specific discriminators, F is the feature extractor, ϑ_f and $\vartheta_{d\text{-intra}}$ are learnable parameters, and e_i indicates whether $\mathbf{x}_i$ is a labeled or unlabeled sample. In this way, we assign the unlabeled target samples to the class domain discriminator using soft labels based on the prediction probabilities, thereby achieving distribution and category matching within the target domain.

Considering supervised learning with target labeled samples, the overall loss of intradomain adversarial learning is as follows:

$$\begin{aligned} \mathcal{L}_{\text{intra}}(\vartheta_f, \vartheta_c, \vartheta_{d\text{-intra}}) &= \mathcal{L}_{c\text{-t}} + \mu \mathcal{L}_{d\text{-intra}} \\ &= \frac{1}{n_{tl}} \sum_{\mathbf{x}_i \in \mathcal{D}_{tl}} \mathcal{L}_c(C(F(\mathbf{x}_i; \vartheta_f); \vartheta_c), y_i) \\ &\quad - \frac{\mu}{n_{tl} + n_{tul}} \sum_{m=1}^M \sum_{\mathbf{x}_i \in \mathcal{D}_t} \mathcal{L}_e(D_{\text{intra}}^m(p_i^m F(\mathbf{x}_i; \vartheta_f); \vartheta_{d\text{-intra}}), e_i). \end{aligned} \quad (2)$$

The adversarial training process is completed by optimizing the following min-max game:

$$\begin{aligned} (\hat{\vartheta}_f, \hat{\vartheta}_c) &= \arg \min_{\vartheta_f, \vartheta_c} \mathcal{L}_{\text{intra}}(\vartheta_f, \vartheta_c, \vartheta_{d\text{-intra}}) \\ (\hat{\vartheta}_{d\text{-intra}}) &= \arg \max_{\vartheta_{d\text{-intra}}} \mathcal{L}_{\text{intra}}(\vartheta_f, \vartheta_c, \vartheta_{d\text{-intra}}). \end{aligned} \quad (3)$$

A gradient reversal layer (GRL) [41] is stacked between F and D_{intra}^m to apply an opposite gradient to the feature extractor during backpropagation

$$\begin{aligned} G_{\text{grl}}(\mathbf{x}) &= \mathbf{x} \\ \frac{d\mathcal{L}_{\text{grl}}}{d\mathbf{x}} &= -\mathbf{I}. \end{aligned} \quad (4)$$

With the GRL, the two distinct optimization problems in (3) can be addressed simultaneously during model training, thereby avoiding separate training

$$\begin{aligned} \vartheta_f &\leftarrow \vartheta_f - \eta \left(\frac{\partial \mathcal{L}_{c\text{-t}}}{\partial \vartheta_f} - \mu \frac{\partial \mathcal{L}_{d\text{-intra}}}{\partial \vartheta_f} \right) \\ \vartheta_c &\leftarrow \vartheta_c - \eta \frac{\partial \mathcal{L}_{c\text{-t}}}{\partial \vartheta_c} \end{aligned}$$

$$\vartheta_{d\text{-intra}} \leftarrow \vartheta_{d\text{-intra}} - \eta \frac{\partial \mathcal{L}_{d\text{-intra}}}{\partial \vartheta_{d\text{-intra}}}. \quad (5)$$

2) *Interdomain Adversarial Adaptation*: To reduce the distribution discrepancy between the source and target domain, we utilize the interdomain loss $\mathcal{L}_{\text{inter}}$ to achieve adversarial training between the feature extractor F and the domain discriminator D_{inter} . In this process, F aims to generate domain-invariant feature representations, while D_{inter} strives to prevent confusion between different domains. The interdomain loss can be formulated as follows:

$$\mathcal{L}_{\text{inter}} = - \frac{1}{n_s + n_t} \sum_{\mathbf{x}_i \in \mathcal{D}_s \cup \mathcal{D}_t} \mathcal{L}_d(D_{\text{inter}}(F(\mathbf{x}_i)), d_i) \quad (6)$$

where d_i indicates whether $\mathbf{x}_i$ is from the source domain or the target domain.

The source classification loss is used to maintain the discriminability of features during the adversarial learning

$$\mathcal{L}_{c\text{-s}} = \frac{1}{n_s} \sum_{\mathbf{x}_i \in \mathcal{D}_s} \mathcal{L}_c(C(F(\mathbf{x}_i)), y_i). \quad (7)$$

Similar training techniques as the intradomain adversarial learning are used in the interdomain adversarial learning.

C. Dynamic Label Adjustment

In the actual adversarial training process, due to the unbalanced learning capabilities of the feature extractor and the discriminators caused by diverse distribution differences, the training is usually unstable [45]. To improve the learning process, we expect to: 1) leverage the available labels on the target well to enrich the supervision information; and 2) focus on more similar source samples during the learning process to enhance the model stability. Inspired by Wang et al. [46], we design a dynamic label adjustment mechanism, which adaptively optimizes the selection of target pseudolabels and relevant source samples, thereby guiding the distribution alignment outside the process of intradomain and interdomain adversarial learning. Moreover, the optimized labels can also reduce the observation bias of the model.

1) *Target Pseudolabeling*: The dynamic pseudolabel adjustment module adaptively adjusts the confidence threshold based on the model's learning state, thereby optimizing the selection of pseudolabels for unlabeled samples. We define a global threshold to control the overall confidence of unlabeled samples, and class-specific thresholds to control fine-grained pseudolabels. In the early iterations, the thresholds are set to low values, allowing more possibly correct samples for training. As the model accuracy improves, the thresholds are dynamically increased to exclude possibly incorrect samples.

More specifically, the global threshold $\mathcal{T}_\tau$ is incrementally adjusted based on the overall learning status, specified as the average prediction confidence, of the model on the target unlabeled samples. The iterative update of $\mathcal{T}_\tau$ is formulated as follows:

$$\mathcal{T}_\tau = \kappa \mathcal{T}_{\tau-1} + (1 - \kappa) \frac{1}{U_\tau} \sum_{u=1}^{U_\tau} \max_m \{p_u^m\}, \tau = 1, 2, \dots \quad (8)$$

where τ denotes the τ th iteration, U_τ denotes the number of unlabeled samples in the τ -iteration, $p_u^m = C(y = m | \mathbf{x}_u)$ is the output probability of the classifier for class m ($\mathbf{x}_u$ is an unlabeled sample), and $\kappa \in (0, 1)$ is the smoothing coefficient. Initially, $\mathcal{T}_0$ is set to $(1/M)$ to accommodate more samples, where M is the number of lithology classes.

To better epitomize the characteristics within each class, we introduce class-specific thresholds $q_\tau(m)$, which are incrementally adjusted based on the model's prediction confidence of a specific class, as follows:

$$q_\tau(m) = \kappa q_{\tau-1}(m) + (1 - \kappa) \frac{1}{U_\tau} \sum_{u=1}^{U_\tau} p_u^m, \tau = 1, 2, \dots \quad (9)$$

Similarly, $q_0(m)$ is initialized to $(1/M)$.

The final dynamic thresholds are obtained by combining the global threshold and the class-specific thresholds, as follows:

$$\mathcal{T}_\tau(m) = \frac{q_{\tau-1}(m) \mathcal{T}_\tau}{\max_m \{q_{\tau-1}(m)\}}. \quad (10)$$

Based on the dynamic thresholds, the target pseudolabeling loss $\mathcal{L}_{tpl}$ for the unlabeled samples is formulated as follows:

$$\mathcal{L}_{tpl} = \frac{1}{U_\tau} \sum_{u=1}^{U_\tau} \mathbb{1} \left(\max_m \{p_u^m\} > \mathcal{T}_\tau \left(\arg \max_m \{p_u^m\} \right) \right) \cdot \mathcal{L}_c(\hat{y}_u, p_u) \quad (11)$$

where $\mathbb{1}(\cdot)$ is the indicator function used for pseudolabel selection, p_u is the predicted probability vector, $\hat{y}_u$ is the one-hot label vector transformed from p_u , and $\mathcal{L}_c$ is the cross-entropy loss.

2) *Source Reweighting*: The dynamic source reweighting module assigns importance weights to source samples according to their similarity with the target samples. In this way, we reshape the distribution of the source domain, thereby alleviating the instability of model adversarial training caused by the great differences between or within domains. Concretely, we pretrain an independent domain discriminator $\tilde{D}$ to distinguish samples from different domains. According to Zhang et al. [47], the optimum domain discriminator is obtained at

$$\tilde{D}(\mathbf{z}) = \frac{\mathcal{P}_t(\mathbf{z})}{\mathcal{P}_s(\mathbf{z}) + \mathcal{P}_t(\mathbf{z})} \quad (12)$$

where $\mathbf{z} = F(\mathbf{x})$. We annotate the source samples as 0 and the target samples as 1 (different from Zhang et al. [47]). The trained $\tilde{D}$ can measure the similarity of each source sample relative to the target domain.

To obtain unbiased estimation between different domains [48], we define the weight of the source sample as the probability density ratio ($\mathcal{P}_t(\mathbf{z})/\mathcal{P}_s(\mathbf{z})$). Then, according to (12), we have

$$w_s(\mathbf{x}) = \frac{\mathcal{P}_t(\mathbf{z})}{\mathcal{P}_s(\mathbf{z})} = \frac{\tilde{D}(F(\mathbf{x}))}{1 - \tilde{D}(F(\mathbf{x}))}. \quad (13)$$

Obviously, (13) increases the weight of similar source samples and reduces the weight of dissimilar ones.

Algorithm 1 : Semi-Supervised Cross-Well Lithology Identification.

Input: Labeled well log data from source well $D_s = \{(\mathbf{x}_i^{tl}, y_i^{tl})\}_{i=1}^{n_{tl}}$; Partially labeled well log data from target well $D_{tl} = \{(\mathbf{x}_i^{tl}, y_i^{tl})\}_{i=1}^{n_{tl}}$ and $D_{tul} = \{\mathbf{x}_i^{tul}\}_{i=1}^{n_{tul}}$.

Output: Lithology predictions for target unlabeled data $\{y_i^{tul}\}_{i=1}^{n_{tul}}$.

- 1: Pre-train an independent domain discriminator $\tilde{D}$ using D_s , D_{tl} , and D_{tul} .
- 2: Input D_s , D_{tl} , and D_{tul} into the model.
- 3: **for** iteration τ from 1 to τ_{max} **do**
- 4: Update the pseudo-labeling threshold $\mathcal{T}_\tau(m)$ using Equation (10).
- 5: Assign pseudo-labels to unlabeled target samples satisfying $\max_m \{p_u^m\} > \mathcal{T}_\tau(\arg \max_m \{p_u^m\})$, and transfer these samples to the labeled target dataset D_{tl} .
- 6: Compute the weight w_s of source samples using Equation (13).
- 7: Optimize the model by minimizing the loss function (16).
- 8: **end for**
- 9: Predict the lithology $\{y_i\}_{i=1}^{n_{tul}}$ of the target unlabeled data using the trained model.

TABLE I
DATASET DESCRIPTION

Dataset 1	Logs ¹		AC, CAL, CNL, GR, R25, SP					
	Lithology ²		MS	SS	OS	DM	Total	
	Sample Size	Well A	2000	385	196	146	2727	
		Well B	2000	435	174	108	2717	
Dataset 2	Logs ¹		AC, CAL, CNL, GR, R25, SP					
	Lithology ²		MS	SI	OS	DM	FS	Total
	Sample Size	Well C	2000	1602	318	105	730	4755
		Well D	2000	1635	244	99	391	4369
		Well E	2000	1395	174	253	150	3972

Note: ¹AC: Acoustic Log, CAL: Caliper Log, CNL: Compensated Neutron Log, GR: Gamma Ray Log, R25: 2.5m Bottom Gradient Resistivity Log, SP: Spontaneous Potential Log. ²MS: Mudstone, SS: Sandstone, SI: Siltstone, OS: Oil Shale, DM: Dolomite, FS: Fine Sandstone.

The weight w_s is introduced into the source-relevant losses $\mathcal{L}_{c \cdot s}$ and $\mathcal{L}_{inter}$ to refine cross-domain feature transfer, as follows:

$$\mathcal{L}_{c \cdot s}^w = \frac{1}{n_s} \sum_{\mathbf{x}_i \in \mathcal{D}_s} w_s(\mathbf{x}_i) \mathcal{L}_c(C(F(\mathbf{x}_i)), y_i) \quad (14)$$

$$\mathcal{L}_{inter}^w = \frac{1}{n_s + n_t} \left(\sum_{i=1}^{n_s} w_s(\mathbf{x}_i) \log(1 - D_{inter}(F(\mathbf{x}_i))) + \sum_{j=1}^{n_t} \log(D_{inter}(F(\mathbf{x}_j))) \right). \quad (15)$$

TABLE II
RECALL (%) OF DIFFERENT METHODS ON WELL A AND WELL B

Method	A→B					B→A				
	MS	SS	OS	DM	Macro-R	MS	SS	OS	DM	Macro-R
SO ¹	60.50	48.87	54.04	40.06	50.87	64.85	62.53	54.98	43.79	56.54
CPM	95.65	50.80	12.04	47.70	51.55	95.50	71.17	11.22	28.08	51.49
ENT	88.22	69.13	47.96	66.24	67.89	88.94	95.58	41.33	70.55	74.10
SSGAN	92.56	68.88	58.16	65.61	71.30	93.90	84.94	70.41	67.12	79.09
LIAFA	89.83	76.53	66.33	57.32	72.50	93.28	89.35	69.90	68.49	80.26
DANN	83.94	76.53	69.67	53.50	70.91	87.00	70.13	40.31	69.86	66.83
MADA	85.50	79.46	78.57	56.05	74.89	<u>96.00</u>	84.42	61.22	69.18	77.71
BSP	80.25	74.02	25.93	41.95	55.54	85.70	70.39	43.88	63.70	65.92
TA	<u>93.50</u>	81.40	60.00	58.82	73.43	94.44	98.44	72.96	76.71	85.64
MME	84.56	79.85	69.39	74.52	77.08	90.94	90.39	84.39	75.34	85.27
UODA	86.67	82.18	78.63	75.07	<u>80.64</u>	97.90	83.12	85.20	88.36	88.65
CDAC	64.33	80.10	66.33	75.16	71.48	77.94	99.41	93.94	<u>94.35</u>	<u>91.41</u>
EFTL	78.11	81.38	46.94	85.99	73.10	63.72	95.10	78.79	95.11	83.18
UniSSDA	76.33	90.56	93.63	54.64	78.79	82.09	94.61	87.88	91.67	89.06
SERL	74.00	84.95	74.49	<u>84.71</u>	79.54	78.28	<u>99.42</u>	98.48	77.40	88.40
SemiDAVIL	92.69	76.54	54.46	64.20	71.97	92.37	99.72	83.82	64.48	85.10
Ours	90.17	<u>87.76</u>	<u>85.35</u>	82.47	86.44	93.50	95.58	<u>95.41</u>	82.19	91.67

Note: ¹SO represents a model trained only on source data and directly applied to target domain lithology prediction. Bold entries indicate the best results. Underlined entries indicate the second-best results.

D. Model Training

Overall, the objective of the proposed semisupervised cross-well lithology identification method is as follows:

$$\mathcal{L} = \alpha \mathcal{L}_{c-s}^w + \beta \mathcal{L}_{\text{inter}}^w + \gamma \mathcal{L}_{\text{intra}} + \lambda \mathcal{L}_{\text{tpl}} \quad (16)$$

where α , β , γ , and λ are hyperparameters used to balance the contributions of each loss term. The training process of the proposed method is summarized in Algorithm 1.

IV. EXPERIMENTS

A. Settings

1) *Data Preparation*: The experimental logging data were collected from an oilfield in eastern China. The experiment utilized six logging curves, which are described in Table I. According to the actual well log data, this section's experiment compiled data from six real exploration wells within the study area into two datasets to comprehensively verify the effectiveness and reliability of the proposed method.

For each logging curve, 1-D image samples are constructed in the depth direction. A certain depth point is taken as the center, and some points are selected both above and below this center. Finally each sample ultimately contains logging data from 33 depth points. This process is repeated sequentially along the depth of the well. Table I shows the lithological categories and sample numbers included in each dataset. Due to the large number of mudstone samples, down-sampling was applied to the mudstone data from all wells, retaining 2000 mudstone image samples to balance the data volume among lithological categories. In the experiment, the labeled target set is constructed using stratified sampling. Specifically, for the target domain, 10% of samples are randomly selected within each class to form the labeled target set, ensuring that all categories are represented. The remaining target samples are treated as unlabeled data. Smaller labeling ratios are further evaluated in

Section IV-B. We use the min-max normalization method to unify logging curves, eliminating systematic errors caused by performance of instruments and measurement technologies.

2) *Evaluation Metrics*: We use recall as the evaluation metric of the model's performance. The formula for calculating recall for a lithology category is defined as follows:

$$\text{Recall}_m = \frac{\text{TP}_m}{\text{TP}_m + \text{FN}_m} \quad (17)$$

where TP_m represents the number of samples that belong to category m and are correctly predicted by the model as category m , and FN_m represents the number of samples that the model fails to correctly predict as category m . To better evaluate the model's comprehensive performance on kinds of categories, the macroaverage recall (macro-R) [49] is used as the evaluation metric. The formula for calculating macro-R is as follows:

$$\text{Macro-R} = \frac{1}{|M|} \sum_{m=1}^M \text{Recall}_m \quad (18)$$

where M represents the total number of lithology categories.

3) *Implementation Details*: To alleviate the impact of class imbalance, we adjust the weights of the supervised loss function based on the number of samples in each category, defined by $\text{weight}_m = N_{\text{total}}/N_m$, where N_m represents the number of samples of a specific lithology type, and N_{total} represents the total number of training samples, ensuring that minority classes receive adequate training. The model is trained using an Adam-based optimizer, with the learning rate set to 0.01 and the weight decay coefficient set to 0.00005 during the training process. An annealing algorithm is employed to adjust the learning rate according to the training progress, ensuring that the model learns quickly in the early stages and converges more stably in the later stages. The hyperparameters are set as $\kappa = 0.9$, $\alpha = 1$, $\beta = 0.2$, $\gamma = 0.5$, and $\lambda = 0.2$. Relevant sensitivity analysis experiments are presented in Section IV-E. All experiments are implemented

TABLE III
RECALL (%) OF DIFFERENT METHODS ON WELL C AND WELL D

Method	C→D						D→C					
	MS	SI	OS	DM	FS	Macro-R	MS	SI	OS	DM	FS	Macro-R
SO	64.40	57.60	58.94	46.64	60.65	60.01	54.85	50.71	31.27	48.70	45.71	46.25
CPM	71.50	64.16	28.69	36.36	71.10	54.36	82.60	75.91	22.37	21.90	47.26	50.01
ENT	87.22	81.04	49.59	40.40	87.98	69.25	90.39	82.02	68.87	62.86	72.88	75.40
SSGAN	93.68	91.44	50.82	44.44	83.63	72.80	90.26	86.45	66.98	60.95	83.84	77.70
LIAFA	96.88	84.77	61.35	56.57	83.30	76.57	93.30	86.70	73.58	71.43	75.62	80.13
DANN	92.33	88.50	54.10	45.45	75.96	71.27	91.50	81.96	70.13	67.62	71.37	76.52
MADA	<u>95.74</u>	81.83	64.34	66.67	77.49	77.21	86.91	85.08	72.01	71.43	77.40	78.57
BSP	92.41	84.89	56.15	41.41	68.29	68.63	84.52	71.97	71.07	61.90	73.01	72.49
TA	93.00	<u>90.46</u>	65.16	64.65	82.35	79.12	95.40	87.02	73.27	70.48	76.03	80.44
MME	93.47	89.72	67.21	60.61	83.12	78.83	90.94	88.01	75.16	73.33	78.90	81.27
UODA	91.48	88.01	70.90	61.62	84.65	<u>79.33</u>	91.54	91.14	77.19	<u>75.24</u>	85.89	<u>84.20</u>
CDAC	76.39	76.36	52.27	90.00	76.14	74.23	72.78	70.94	81.18	69.47	<u>89.04</u>	76.68
EFTL	82.78	64.54	68.64	50.00	90.34	71.26	74.72	77.74	<u>85.37</u>	70.53	93.00	80.27
UniSSDA	81.44	65.69	<u>75.00</u>	<u>82.02</u>	86.93	78.22	71.84	80.95	89.73	65.12	81.29	77.79
SERL	66.56	70.31	63.64	81.11	98.86	76.10	80.11	90.50	64.81	87.37	80.82	80.72
SemiDAViL	87.53	83.30	35.24	68.82	<u>92.58</u>	73.49	90.32	<u>92.01</u>	54.39	55.10	82.33	74.83
Ours	93.72	88.32	80.45	67.42	84.10	82.80	<u>95.35</u>	92.82	79.87	70.48	83.15	84.33

Note: Bold entries indicate the best results. Underlined entries indicate the second-best results.

TABLE IV
RECALL (%) OF DIFFERENT METHODS ON WELL C AND WELL E

Method	C→E						E→C					
	MS	SI	OS	DM	FS	Macro-R	MS	SI	OS	DM	FS	Macro-R
SO	69.16	62.64	54.10	44.96	60.29	58.23	51.54	39.00	42.19	49.81	51.37	46.78
CPM	60.10	77.92	32.18	33.20	28.67	46.41	87.25	62.92	34.29	28.57	32.33	49.07
ENT	85.99	84.09	67.82	72.33	62.00	74.45	86.78	83.02	68.87	62.86	72.05	74.72
SSGAN	87.17	84.80	64.37	69.96	72.00	75.66	90.92	84.96	72.96	60.95	76.03	77.16
LIAFA	89.26	<u>88.53</u>	69.54	71.94	74.67	78.79	93.30	86.02	73.58	71.43	78.77	80.62
DANN	82.56	86.74	68.39	69.17	70.00	75.37	88.11	72.03	71.07	69.52	71.51	74.45
MADA	86.23	82.94	72.99	73.91	71.33	77.48	89.06	81.96	75.16	71.43	81.37	79.80
BSP	84.35	80.00	64.37	65.22	63.33	71.45	87.05	75.59	68.24	62.86	65.07	71.76
TA	<u>91.40</u>	85.81	70.69	71.94	68.67	77.70	91.70	82.02	71.70	70.48	73.01	77.78
MME	90.09	87.17	69.54	72.73	74.00	78.71	93.78	83.96	70.44	72.38	77.12	79.54
UODA	91.32	89.24	76.85	75.10	84.67	<u>83.44</u>	91.28	84.96	78.93	73.33	80.00	<u>81.70</u>
CDAC	81.67	73.01	71.97	<u>81.58</u>	74.07	76.46	72.89	87.59	86.45	63.16	75.96	77.21
EFTL	78.44	59.39	<u>81.53</u>	56.14	96.30	74.36	80.06	72.19	<u>97.26</u>	42.11	72.47	72.82
UniSSDA	79.47	78.43	82.72	80.37	89.04	82.01	68.51	85.00	81.58	74.42	<u>81.51</u>	78.20
SERL	79.33	70.86	78.34	87.28	76.30	78.42	71.33	81.28	99.09	88.42	65.85	81.19
SemiDAViL	87.47	86.52	69.75	56.36	76.43	75.31	82.63	<u>90.34</u>	78.50	42.86	63.51	71.57
Ours	91.48	90.18	77.59	75.89	<u>89.33</u>	84.89	<u>93.72</u>	91.05	73.87	<u>75.79</u>	85.84	84.05

Note: Bold entries indicate the best results. Underlined entries indicate the second-best results.

on a computer server with an NVIDIA GeForce RTX 3090 GPU.

B. Comparison Study

In this section, we compare our proposed method with 15 baseline methods from four related research directions to validate the effectiveness of the method presented in this article. These baseline methods include traditional lithology identification methods (CPM), semisupervised learning methods (ENT [50], SSGAN [51], LIAFA [52]), unsupervised domain adaptation methods (DANN [41], MADA [53], BSP [54]), and semisupervised domain adaptation methods (TA [55], MME [56], UODA [57], CDAC [33], EFTL [36], UniSSDA [38], SERL [37], SemiDAViL [58]).

We conducted experiments on different source-target combinations. The results are shown in Tables II–V. We have the following observations: 1) the proposed method exhibits advantages across all testing scenarios. Compared with the traditional lithology identification method (CPM) and the best-performing semisupervised learning (LIAFA), unsupervised domain adaptation (MADA), and semisupervised domain adaptation methods (UODA), our method achieves average improvements of 33.3%, 6.9%, 8.0%, and 2.2% in macro-R across all six combinations, respectively, with an overall average of 12.6%; 2) although the proposed model does not achieve the best performance on each individual class, it produces a more balanced per-class prediction accuracy compared to the baselines, which can be attributed to the class-specific discrimination

TABLE V
RECALL (%) OF DIFFERENT METHODS ON WELL D AND WELL E

Method	D→E						E→D					
	MS	SI	OS	DM	FS	Macro-R	MS	SI	OS	DM	FS	Macro-R
SO	74.23	66.46	54.59	56.24	51.34	60.57	72.66	64.27	59.30	50.04	60.18	61.29
CPM	94.35	70.90	39.20	47.91	50.67	60.61	73.40	71.74	52.87	39.39	54.48	58.38
ENT	92.11	87.89	67.24	72.73	77.33	79.46	87.50	83.12	62.30	63.64	73.40	74.00
SSGAN	88.93	85.95	78.16	75.49	85.33	82.77	91.76	82.02	68.44	67.68	78.01	77.58
LIAFA	89.26	88.03	81.03	79.05	84.67	84.41	92.88	83.98	71.31	68.69	83.12	79.99
DANN	<u>96.30</u>	90.97	66.67	70.36	66.00	78.06	82.17	80.00	66.80	66.67	72.12	73.55
MADA	97.70	89.03	72.41	65.22	76.67	80.21	90.00	84.04	74.18	69.70	74.94	78.57
BSP	95.33	76.99	66.09	73.91	72.67	76.99	80.65	71.99	66.39	61.62	71.10	70.35
TA	92.50	86.95	80.46	81.03	78.00	83.79	<u>92.20</u>	<u>86.97</u>	69.67	60.61	79.03	77.70
MME	92.94	88.46	<u>86.78</u>	83.00	81.33	86.50	92.05	83.98	72.95	68.69	73.66	78.27
UODA	93.33	89.03	83.91	82.61	88.00	<u>87.38</u>	91.17	85.99	78.69	73.74	82.10	82.34
CDAC	83.56	66.88	56.69	70.61	<u>94.81</u>	74.51	56.22	73.23	78.18	<u>88.89</u>	86.65	76.63
EFTL	76.78	73.96	85.93	57.96	71.49	73.22	78.50	67.26	62.27	66.67	<u>91.48</u>	73.24
UniSSDA	83.68	73.64	82.71	75.70	97.26	82.60	75.17	73.41	80.34	84.21	89.35	80.50
SERL	78.61	81.05	92.59	<u>87.72</u>	69.43	81.88	65.72	75.00	84.55	93.33	97.73	<u>83.27</u>
SemiDAViL	85.38	92.91	70.99	61.02	65.71	75.20	88.82	86.46	54.19	44.09	82.14	71.14
Ours	95.54	<u>92.33</u>	81.03	90.51	91.33	90.15	91.33	87.22	<u>82.38</u>	72.73	86.19	83.97

Note: Bold entries indicate the best results. Underlined entries indicate the second-best results.

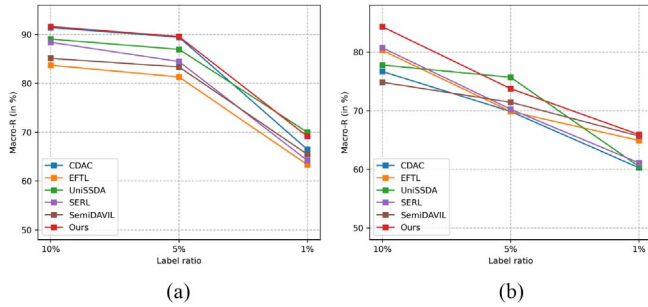


Fig. 3. Comparison of model performance under different target label ratios. (a) B→A. (b) D→C.

strategy and class-specific pseudolabel thresholds adopted in the model; and 3) semisupervised domain adaptation methods outperform semisupervised methods, which in turn outperform unsupervised domain adaptation methods. This reflects the positive impact of incorporating target domain label information. Semisupervised domain adaptation methods effectively leverage large amounts of source domain data and a small amount of labeled target domain data, excelling in knowledge transfer and feature alignment.

To evaluate the stability of our method under different target label ratios, we present the experimental results using 10%, 5%, and 1% labeled target data, as shown in Fig. 3. As the label ratio decreases, the performance of all methods declines. Our method exhibits a more gradual performance degradation and achieves the best or second-best performance across all label ratios, demonstrating its stability under limited supervision.

Fig. 4 compares the inference time and number of model parameters of different methods. Our method exhibits moderate model complexity, outperforming several recent semisupervised domain adaptation methods (EFTL, UniSSDA, SERL, and SemiDAViL), while being slightly less efficient than some earlier methods. TA has the lowest model complexity, as it

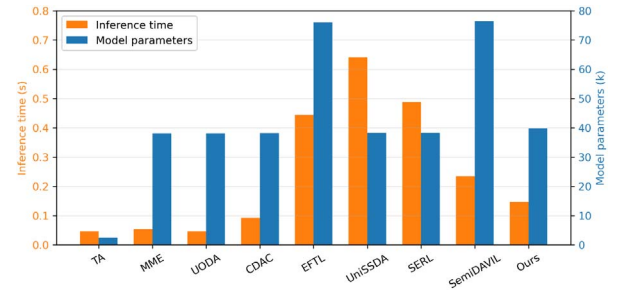


Fig. 4. Comparison of inference time and number of model parameters.

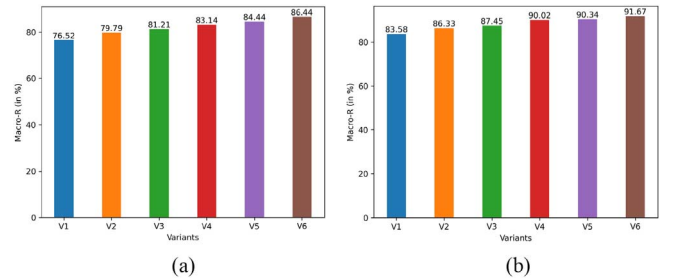


Fig. 5. Results of the model ablation study. (a) A→B. (b) B→A.

is based on an ensemble learning framework rather than a deep neural network architecture. Overall, our method achieves a favorable trade-off between model complexity and performance.

C. Ablation Study

This section conducts a series of ablation studies by sequentially removing or replacing specific modules to analyze their impact on model performance.

- 1) V_6 : The complete structure of the proposed model.

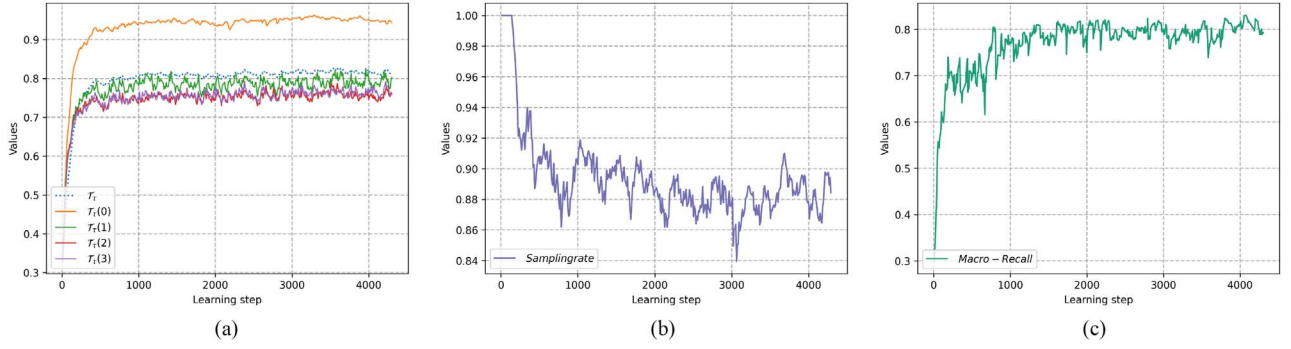


Fig. 6. Dynamic adjustment process of global and class-specific thresholds. (a) Threshold adjustments. (b) Sampling rate. (c) Macro-R.

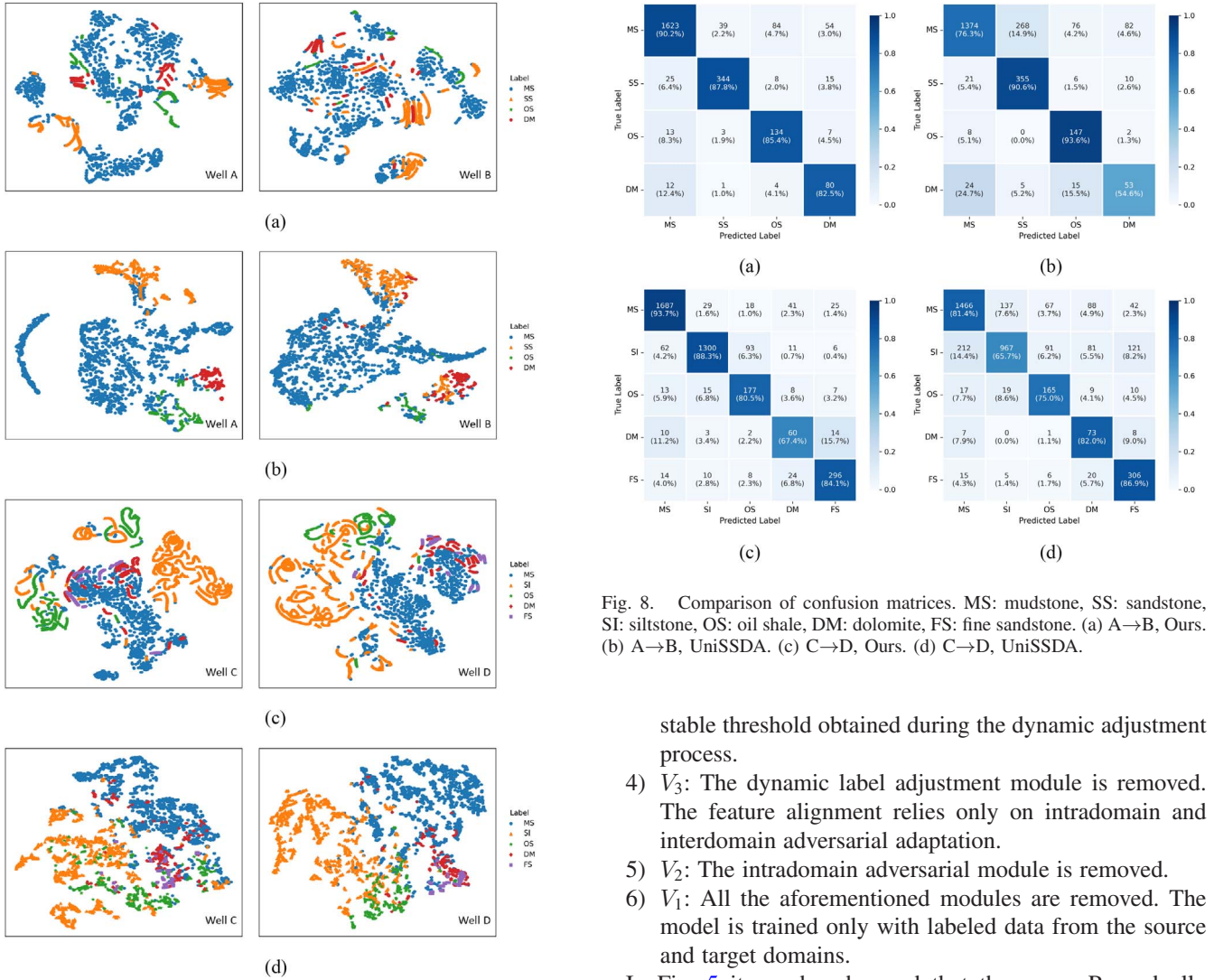


Fig. 8. Comparison of confusion matrices. MS: mudstone, SS: sandstone, SI: siltstone, OS: oil shale, DM: dolomite, FS: fine sandstone. (a) A→B, Ours. (b) A→B, UniSSDA. (c) C→D, Ours. (d) C→D, UniSSDA.

Fig. 7. t-SNE visualization of the input space and feature space. MS: mudstone, SS: sandstone, SI: siltstone, OS: oil shale, DM: dolomite, FS: fine sandstone. (a) Original data distribution of wells A and B. (b) Feature space distribution of wells A and B. (c) Original data distribution of wells C and D. (d) Feature space distribution of wells C and D.

- 2) V_5 : The source reweighting module is removed.
- 3) V_4 : The target pseudolabeling module is removed, and a fixed threshold is set for each class based on the final

stable threshold obtained during the dynamic adjustment process.

- 4) V_3 : The dynamic label adjustment module is removed. The feature alignment relies only on intradomain and interdomain adversarial adaptation.
- 5) V_2 : The intradomain adversarial module is removed.
- 6) V_1 : All the aforementioned modules are removed. The model is trained only with labeled data from the source and target domains.

In Fig. 5 it can be observed that the macro-R gradually increases as more model components are added, which verifies the irreplaceable role of each component in enhancing model performance. The macro-R values for the base model V_1 are 76.52% and 83.58%, which are the lowest. This indicates that relying solely on basic classification functions is insufficient to capture complex features. V_2 improves the macro-R values to 79.79% and 86.33%. This highlights the importance of considering the distribution discrepancy between domains. s

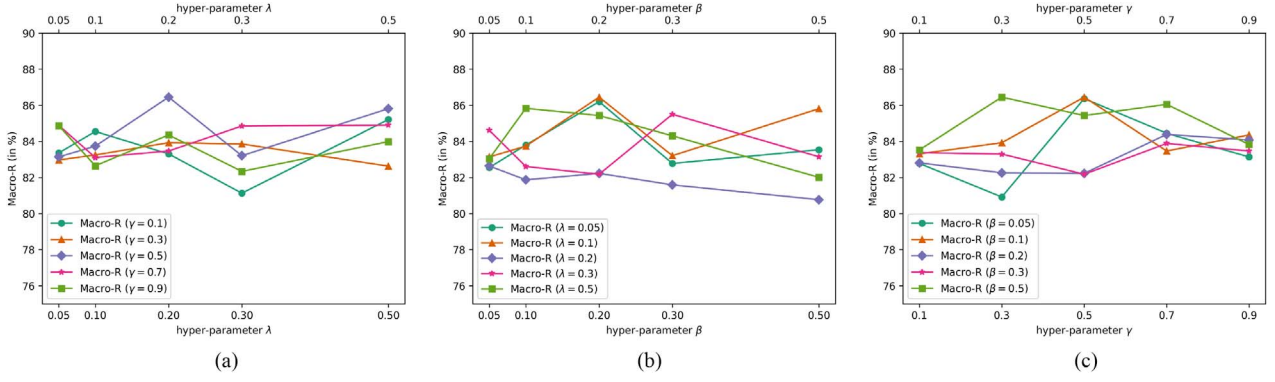


Fig. 9. Sensitivity analysis of model hyperparameters. (a) $\beta = 0.2$. (b) $\gamma = 0.5$. (c) $\lambda = 0.2$.

for V_3 , the macro-R value results are higher than those of V_2 . This indicates the effectiveness of the intradomain adversarial module in lithology prediction. V_4 removes the dynamic label adjustment modules compared to V_6 . The results indicate that the threshold dynamic adjustment module plays a key role in addressing the issue of imbalanced samples. The results of V_5 validate the effectiveness of the source reweighting. The complete model V_6 exhibits the best performance. This indicates that the combination of all components significantly enhances the overall performance of the model.

D. Visualization Analysis

To more intuitively demonstrate the dynamic threshold adjustment process and its effect, we record the global and class-specific thresholds and sampling rate of pseudolabels during the iterative training process, and also plot the performance curve in terms of macro-R. In Fig. 6(a) and (b), the global threshold (i.e. $\mathcal{T}_\tau$) and class-specific thresholds (i.e. $\mathcal{T}_\tau(m)$) are moderately low at the early stage, leading to a high utilization rate of pseudolabels (as observed through the sampling rate), which in turn accelerates convergence. Subsequently, the thresholds gradually increase during training and eventually stabilize at a high level, indicating that the model's confidence in its predictions is progressively enhanced, thereby reducing confirmation bias. Accordingly, the model's classification performance is also improved, as demonstrated by the macro-R in Fig. 6(c).

Additionally, we use t-SNE [59] to visualize the distribution of logging data and feature space. It can be observed that there are significant discrepancies in the distribution of logging data in the original input space [Fig. 7(a) and (c)]. Moreover, samples belonging to the same lithology category are often scattered across multiple clusters, reflecting the intrawell data divergence. Fig. 7(b) and (d) shows the distributions of different wells in the feature space learned by the proposed method. Compared with the original input space, the distributions of different wells become more consistent, and samples from the same category tend to cluster together. This demonstrates that the proposed method is capable of extracting domain-invariant and compact features from the well log data.

To analyze which classes are prone to misclassification, we plot the confusion matrices (as shown in Fig. 8), and compare our method with the baseline UniSSDA, which performs relatively well on certain classes. Our method shows slightly inferior performance to UniSSDA in identifying oil shale (OS) and dolomite (DM), while achieving substantially better performance on the other lithologies. Overall, our method demonstrates a more balanced classification performance across different classes. From Fig. 8(a) and (c), we notice that our method encounters difficulty in identifying DM. Specifically, on $A \rightarrow B$, 12.4% of DM samples are misclassified as mudstone (MS), while on $C \rightarrow D$, 11.2% and 15.7% of DM samples are misclassified as MS and fine sandstone (FS), respectively. The reasons are twofold: 1) lithologies usually exhibit strong heterogeneity, and the logging responses of DM overlap with those of mud-rich facies and clean clastic rocks; and 2) MS has the largest number of training samples among all classes, which may bias the model toward learning MS-related features. In future work, we will incorporate class-imbalance-aware designs into the proposed method to further enhance the identification performance on minority lithologies.

E. Sensitivity Analysis

The evaluation of the sensitivity of the model to key hyperparameters shows robustness of our method. For convenience, we only present the macro-R values for the lithology prediction of well A $\rightarrow$ B.

We study the hyperparameters β , γ , and λ in the model's loss function (16). As shown in Fig. 9, the results of different combinations show that the fluctuations in the experimental results due to different weight of loss function are not significant. The parameter β corresponds to the weight of the interdomain adversarial adaptation loss. As it increases, the recall first increases and then decreases, with its optimal setting being 0.5. The parameter γ corresponds to the weight of the intradomain adversarial adaptation loss. As it increases, the recall also initially increases before decreasing, with its optimal setting being 0.5. The parameter λ corresponds to the weight of the target pseudolabeling loss. As it increases, the recall first decreases and then increases, with its optimal setting being 0.2.

Additionally, it can be observed that adjustments to these parameters cause only minor fluctuations in model performance, thus confirming the robustness of our method.

V. CONCLUSION

In this article, we propose a dual adversarial adaptation method with dynamic labeling mechanism for semisupervised cross-well lithology identification. Specifically, we introduce a dual adversarial adaptation method, incorporating adversarial adaptation both within and across domains, to mitigate intrawell divergence and interwell discrepancies. To fully leverage the target label information and improve the adversarial learning process, we propose a dynamic label adjustment mechanism to select reliable target pseudolabeled samples and informative source samples, thereby facilitating feature alignment and training convergence. Experimental results on a real-world well log datasets demonstrate that our method outperforms ten baseline methods from multiple related fields when target labels are scarce. Ablation study and sensitivity analysis validate the rationality of the module design in our method. Moreover, visualization of intermediate features shows that our method is capable of learning domain-invariant and compact feature representations, thereby improving lithology identification across wells.

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